C4.2 Transfers of energy and matter

NAME	DATE	TOTAL
		_____ / 52

About 55 minutes, or one section at a time. You will need a calculator. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 DISTINGUISH between open and closed systems. [C4.2.1] [2]

2 STATE the main source of energy for most ecosystems. [C4.2.2] [1]

3 NAME one chemoautotroph. [C4.2.7] [1]

4 LIST three sources of energy for decomposers. [C4.2.5] [2]

5 STATE the units used for primary production. [C4.2.15] [1]

6 LIST three fuels whose combustion releases carbon dioxide. [C4.2.19] [3]

Section B: Understanding

18 MARKS

7 EXPLAIN why energy cannot be recycled, whereas matter can. [C4.2.13, C4.2.22] [3]

8 COMPARE AND CONTRAST photoautotrophs and chemoautotrophs. [C4.2.6, C4.2.7] [3]

PHOTOAUTOTROPHS	CHEMOAUTOTROPHS

9 **EXPLAIN** why there are rarely more than four or five trophic levels. [C4.2.12, C4.2.14] [3]

10 **EXPLAIN** why secondary production is lower than primary production. [C4.2.16] [3]

11 **EXPLAIN** how an ecosystem can change from a carbon sink to a carbon source. [C4.2.18] [3]

12 **OUTLINE** the interdependence of aerobic respiration and photosynthesis. [C4.2.21] [3]

Section C: Working with data

16 MARKS

Data in question 13 are invented for practice. Question 14 follows the real pattern at Mauna Loa, with rounded figures.

13 [C4.2.10–12]

Energy gained per year in a pond was measured, in kJ m^{-2} yr^{-1}.

GROUP	ENERGY
ALGAE	40000
WATER FLEAS	5000
SMALL FISH	450
HERON	30

a DRAW a pyramid of energy for this pond, with labels. [3]

b CALCULATE the percentage of energy passed from algae to water fleas. [1]

c SUGGEST two reasons why not all energy reached the small fish. [2]

d STATE the trophic level of small fish. [1]

14 [C4.2.20]

YEAR	1960	1980	2000	2020
YEARLY MEAN CARBON DIOXIDE / PPM	317	339	370	414

a CALCULATE the mean rate of rise, in ppm per year, between 1960 and 2020. [2]

b DESCRIBE how the rate of rise changes. [1]

c EXPLAIN the long-term rise. [2]

d Each year, carbon dioxide also rises and falls by about 6 ppm. EXPLAIN this pattern. [4]

Section D: Longer answer

8 MARKS

15 DRAW a labelled carbon cycle, and **explain** how burning fossil fuels affects it. [C4.2.17, C4.2.19] [8]